RYAN J. ECKERT, PH.D.

Coral reef ecology, mesophotic coral ecosystems, molecular ecology



EDUCATION

2025 | 2019
Ph.D. in Integrative Biology
Florida Atlantic University — Boca Raton, FL
Dissertation: Population genetics of shallow and mesophotic corals and sponges in Florida and the Gulf of Mexico
Advisor: Dr. Joshua Voss

2019 | 2017
M.Sc. in Biological Sciences
Florida Atlantic University — Boca Raton, FL
Thesis: Population genetics and algal symbiont communities of the coral species *Montastraea cavernosa* on the Belize Barrier Reef
Advisor: Dr. Joshua Voss

2009 | 2005
B.Sc. in Biological Sciences
Florida State University — Tallahassee, FL
Thesis: Associational resistance and susceptibility in *Solanum carolinense*
Advisor: Dr. Nora Underwood

RESEARCH EXPERIENCE

Present | 2025
Postdoctoral Scholar
FAU, Harbor Branch Oceanographic Institute — Fort Pierce, FL
Coral Reef Health and Molecular Ecology Lab

2025 | 2017
Graduate Research Assistant
FAU, Harbor Branch Oceanographic Institute — Fort Pierce, FL
Coral Reef Health and Molecular Ecology Lab

2017 | 2009
Research Specialist
NOAA's Flower Garden Banks NMS — Galveston, TX
Science Division

2009 | 2007
Research Assistant
Florida State University — Tallahassee, FL
Underwood Lab

TEACHING EXPERIENCE

2022
Graduate Teaching Assistant / Lecturer
FAU / FIO — Long Key, FL
FIO Summer Marine Field Studies

2017
Graduate Teaching Assistant
FAU, Harbor Branch Oceanographic Institute — Fort Pierce, FL
Marine Biodiversity

CONTACT

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ryan.j.eckert@gmail.com
ryanjeckert.weebly.com
cnidaryan.bsky.social
RyanEckert
Ryan J Eckert

SKILLS

Coral reef ecology
Technical SCUBA diving
Closed-circuit rebreather diving
1000+ Scientific Dives
500+ days at sea
100+ research expeditions
Multivariate data analysis
NGS library prep
Bioinformatic analyses

LANGUAGE SKILLS

R
Bash
Python

2014 | 2013

Guest Lecturer
Texas A&M University at Galveston — Galveston, TX
Scientific Diving Field Survey Methods

2010

Instructor
Galveston Independent School District — Galveston, TX
21st Century ACE

STUDENTS MENTORED

2018

Emily Chei
Link fellowship intern
Assessing population connectivity of the coral *Montastraea cavernosa* along the northern Florida Reef Tract
Presently a Ph.D. student at The University of Hong Kong

2017

Ashley Reaume
Link fellowship intern
Optimized extraction and amplification of algal symbiont DNA from TRIzol preserved corals
Presently a Ph.D. student at The University of Central Florida

REFEREED PUBLICATIONS

2025

26. Genetic lineages do not predict stony coral tissue loss disease resistance or survivorship among coral restoration outplants in Florida
Restor Ecol (2025). doi: 10.1111/rec.70234
Bell SL, Shilling EN, Pantoni GS, Eckert RJ, Sturm AB, Sharp WC, Muller EM, Gilliam DS, Lirman D, Bourque AS, Voss JD

2025

25. Benthic community change and stress tolerant coral at a high-latitude coral community in the northwestern Gulf of Mexico
Coral Reefs (2025). doi: 10.1007/s00338-025-02737-3
Nuttall MF, O'Connell K, Eckert RJ, Hickerson EL, Eisenbach O, French D, DeBose JL, Johnston MA, Schmahl GP

2024

24. Recent trends and biases in mesophotic ecosystem research
Biol Lett (2024). doi: 10.1098/rsbl.2024.0465
Radice VZ, Hernandez-Agreda A, Pérez-Rosales G, Booker R, Bellworthy J, Broadribb M, Carpenter GE, Diaz C, Eckert RJ, Foster NL, Gijsbers JC, Gress E, Laverick JH, Micaroni V, Pierotti M, Rouzé H, Stevenson A, Sturm AB, Bongaerts P

2024

23. Assesment of nutrient amendments on stony coral tissue loss disease in Southeast Florida
Front Mar Sci (2024). doi: 10.3389/fmars.2024.1384534
Carreiro AM, Eckert RJ, Sturm AB, Ingalls TC, Combs IR, Walker BK, Voss JD

2024

22. Cryptic diversity of shallow and mesophotic Stephanocoenia intersepta corals across Florida Keys National Marine Sanctuary
Heredity (2024). doi:10.1038/s41437-024-00698-x
Eckert RJ, Sturm AB, Carreiro AM, Klein AM, Voss JD

Citations: 876

h-index: 18

i10-index: 26

2024

21. Algal symbiont genera but not coral host genotypes correlate to stony coral tissue loss disease susceptibility among *Orbicella faveolata* colonies in South Florida
Front Mar Sci (2024). doi:10.3389/fmars.2024.1287457
Klein, AM, Sturm AB, Eckert RJ, Walker BK, Neely KL, Voss JD

2023

20. Does depth divide? Variable genetic connectivity patterns among shallow and mesophotic *Montastraea cavernosa* coral populations across the Gulf of Mexico and western Caribbean
Ecol Evol (2023). doi:10.1002/ece3.10622
Sturm AB, Eckert RJ, Carreiro AM, Klein, AM, Studivan MS, Dodge Farelli D, Simões N; González-Díaz P, González Méndez J, Voss JD

2023

19. *Porites astreoides* coral populations demonstrate high clonality and connectivity in southeast Florida
Coral Reefs (2023). doi:10.1007/s00338-023-02417-0
Shilling EN, Eckert RJ, Sturm AB, Voss JD

2023

18. Stony coral tissue loss disease intervention with amoxicillin leads to a reversal of disease-modulated gene expression pathways
Mol Ecol (2023). doi:10.1111/mec.17110
Studivan MS, Eckert RJ, Shilling EN, Soderberg N, Enochs IC, Voss JD

2023

17. Diet and Feeding Ecology of the Whitespotted Eagle Ray (*Aetobatus narinari*) from Florida Coastal Waters Revealed via DNA Barcoding
Fishes (2023). doi:10.3390/fishes8080388
Cahill BV, Eckert RJ, Bassos-Hull K, Ostendorf TJ, Voss JD, DeGroot BC, Ajemian MJ

2023

16. Coral disease outbreak at the remote Flower Garden Banks, Gulf of Mexico
Front Mar Sci (2023). doi:10.3389/fmars.2023.1111749
Johnston MA, Studivan MS, Enochs IC, Correa AMS, Besemer N, Eckert RJ, Edwards K, Hannum R, Hu X, Nuttall MF, O'Connel K, Palacio-Castro AM, Schmahl GP, Sturm AB, Ushijima B, Voss JD

2022

15. Depth-dependent genetic structuring of a depth-generalist coral and its Symbiodiniaceae algal communities at Campeche Bank, Mexico
Front Mar Sci (2022). doi:10.3389/fmars.2022.835789
Sturm AB, Eckert RJ, Carreiro AM, Simões N, Voss JD

2022

14. Population genetic structure of a broadcast spawning coral, *Montastraea cavernosa*, demonstrates refugia potential of upper mesophotic populations in the Florida Keys
Coral Reefs (2022). doi:10.1007/s00338-021-02112-y
Sturm AB, Eckert RJ, Carreiro AM, Voss JD

2021

13. Quantifying impacts of stony coral tissue loss disease on corals in Southeast Florida through surveys and 3D photogrammetry
PLoS ONE (2021). doi:10.1371/journal.pone.0252593
Combs IR, Studivan MS, Eckert RJ, Voss JD

2020

12. Population genetic structure of the great star coral, *Montastraea cavernosa*, across the Cuban archipelago with comparisons between microsatellite and SNP markers

Sci Rep (2020). doi:10.1038/s41598-020-72112-5

Sturm AB, Eckert RJ, González Méndez J, González-Díaz P, Voss JD

2020

11. Depth influences Symbiodiniaceae associations among *Montastraea cavernosa* corals on the Belize Barrier Reef

Front Microbiol (2020). doi:10.3389/fmicb.2020.00518

Eckert RJ, Reaume AM, Sturm AB, Studivan MS, Voss JD

2020

10. Population structure of the scleractinian coral, *Montastraea cavernosa*, in Southeast Florida

Bull Mar Sci (2020). doi:10.5343/bms.2019.0074

Dodge DL, Studivan MS, Eckert RJ, Chei E, Beal J, Voss JD

2019

9. Populations of the coral species *Montastraea cavernosa* on the Belize Barrier Reef lack vertical connectivity

Sci Rep (2019). doi:10.1038/s41598-019-43479-x

Eckert RJ, Studivan MS, Voss JD

2019

8. The Mesoamerican Reef

In: Loya Y, Puglise KA, Bridge TCL (eds) Mesophotic Coral Ecosystems of the World. Springer International Publishing, Cham, pp 71–84 (2019). doi:10.1007/978-3-319-92735-0_5

Gress E, Voss JD, Eckert RJ, Rowlands G, Andradi-Brown DA

2019

7. Coral bleaching and recovery from 2016 to 2017 at East and West Flower Garden Banks, Gulf of Mexico

Coral Reefs (2019). doi:10.1007/s00338-019-01788-7

Johnston MA, Hickerson EL, Nuttall MF, Blakeway RD, Sterne TK, Eckert RJ, Schmahl GP

2019

6. Localized coral reef mortality event at East Flower Garden Bank, Gulf of Mexico

Bull Mar Sci (2019). doi:10.5343/bms.2018.0057

Johnston MA, Nuttall MF, Eckert RJ, Blakeway RD, Sterne TK, Hickerson EL, Schmahl GP, Lee MT, MacMillan J, Embesi JA

2018

5. Seasonal variability of carbonate chemistry and decadal changes in waters of a marine sanctuary in the Northwestern Gulf of Mexico

Mar Chem (2018). doi:10.1016/j.marchem.2018.07.006

Hu X, Nuttall MF, Wang H, Yao H, Staryk CJ, McCutcheon MR, Eckert RJ, Embesi JA, Johnston MA, Hickerson EL, Schmahl GP, Manzello D, Enochs IC, DiMarco S, Barbero L

2016

4. Persistence of coral assemblages at East and West Flower Garden Banks, Gulf of Mexico

Coral Reefs (2016). doi:10.1007/s00338-016-1452-x

Johnston MA, Embesi JA, Eckert RJ, Nuttall MF, Hickerson EL, Schmahl GP

2016

3. Rapid invasion of Indo-Pacific lionfishes *Pterois volitans* (Linnaeus, 1758) and *P. miles* (Bennett, 1828) in Flower Garden Banks National Marine Sanctuary, Gulf of Mexico, documented in multiple data sets

BioInvasions Rec (2016). doi:10.3391/bir.2016.5.2.09

Johnston MA, Nuttall MF, Eckert RJ, Embesi JA, Sterne TK, Hickerson EL, Schmahl GP

2014

2. Lionfish (*Pterois volitans* [Linnaeus, 1758] and *P. miles* [Bennett, 1828]) records within mesophotic depth ranges on natural banks in the Northwestern Gulf of Mexico

BioInvasions Rec (2014). doi:10.3391/bir.2014.3.2.09

Nuttall MF, Johnston MA, Eckert RJ, Embesi JA, Hickerson EL, Schmahl GP

2012

1. Flower Garden Banks – A refuge in the Gulf of Mexico?

In: Yellowlees D, Hughes TP (eds) Proceedings of the 12th International Coral Reef Symposium (2012).

Hickerson EL, Schmahl GP, Johnston MA, Nuttall MF, Embesi JA, Eckert RJ

MANUSCRIPTS IN PREPARATION

Population genetics of the giant barrel sponge, *Xestospongia muta*, reveal distinct, hybridizing lineages across the Florida Reef Tract (*in review*)

Eckert RJ, Sturm AB, Carreiro AM, Voss JD

Genetic insights of key habitat-forming coral and sponge populations in Flower Garden Banks National Marine Sanctuary (*in prep*)

Eckert RJ, Sturm AB, Pantoni GS, Herrera S, Voss JD

TECHNICAL REPORTS

2018

Stetson Bank Long-term monitoring: 2016 annual report

Marine Sanctuaries Conservation Series ONMS-19-02. U.S. Department of Commerce, NOAA, FGBNMS, Galveston, TX. (2018).

Nuttall MF, Sterne TK, Eckert RJ, Hu X, Sinclair J, Hickerson EL, Embesi JA, Johnston MA, Schmahl GP

2017

Long-term monitoring at East and West Flower Garden Banks: 2016 annual report

Marine Sanctuaries Conservation Series ONMS-17-09. U.S. Department of Commerce, NOAA, FGBNMS, Galveston, TX. (2017).

Johnston MA, Sterne TK, Eckert RJ, Nuttall MF, Embesi JA, Walker RD, Hu X, Hickerson EL, Schmahl GP

2017

Long-term monitoring at Stetson Bank, Flower Garden Banks National Marine Sanctuary, 2015 annual report

Marine Sanctuaries Conservation Series ONMS-17- 06. U.S. Department of Commerce, NOAA, FGBNMS, Galveston, TX. (2017).

Johnston MA, Sterne TK, Eckert RJ, Nuttall MF, Embesi JA, Walker RD, Hu X, Hickerson EL, Schmahl GP

2017 **Long-term monitoring at the East and West Flower Garden Banks National Marine Sanctuary, 2013–2015, volume 1: technical report**
U.S. Dept. of Interior, Bureau of Ocean Energy Management, Gulf of Mexico OCS Region, New Orleans, LA. OCS Study BOEM 2017-058. (2017).
Johnston MA, Eckert RJ, Nuttall MF, Sterne TK, Embesi JA, Manzello DP, Hickerson EL, Schmahl GP

2016 **Long-term monitoring at East and West Flower Garden Banks: 2015 annual report**
U.S. Marine Sanctuaries Conservation Series ONMS-16-02. U.S. Department of Commerce, National Oceanic and Atmospheric Administration, Flower Garden Banks National Marine Sanctuary, Galveston, TX. (2016).
Johnston MA, Eckert RJ, Sterne TK, Nuttall MF, Hu X, Embesi JA, Hickerson EL, Schmahl GP

2015 **Long-term monitoring at East and West Flower Garden Banks National Marine Sanctuary: 2014 annual report**
U.S. Department of Commerce, NOAA, FGBNMS, Galveston, TX. (2015).
Johnston MA, Sterne TK, Eckert RJ, Nuttall MF, Embesi JA, Walker RD, Hu X, Hickerson EL, Schmahl GP

2015 **Long-term monitoring at East and West Flower Garden Banks National Marine Sanctuary, 2011–2012, volume 1: technical report**
U.S. Dept. of Interior, Bureau of Ocean Energy Management, Gulf of Mexico OCS Region, New Orleans, Louisiana. OCS Study BOEM 2015-027. (2015).
Johnston MA, Nuttall MF, Eckert RJ, Embesi JA, Slowey NC, Hickerson EL, Schmahl GP

2014 **Long-term monitoring at East and West Flower Garden Banks National Marine Sanctuary: 2013 annual report**
U.S. Department of Commerce, NOAA, FGBNMS, Galveston, TX. (2014).
Johnston MA, Nuttall MF, Eckert RJ, Embesi JA.

2013 **Long-term monitoring at the East and West Flower Garden Banks National Marine Sanctuary, 2009–2010, volume 1: technical report**
U.S. Dept. of Interior, Bureau of Ocean Energy Management, Gulf of Mexico OCS Region, New Orleans, Louisiana. OCS Study BOEM 2013-214. (2013).
Johnston MA, Nuttall MF, Eckert RJ, Embesi JA, Slowey NC, Hickerson EL, Schmahl GP

PRESENTATIONS

2026 **Population genetic structuring varies across depth and region in corals and sponges in marine protected areas of the Gulf of Mexico and Florida**
16th International Coral Reef Symposium — Auckland, NZ
Eckert RJ, Sturm AB, Carreiro AM, Pantoni GS, Voss JD
Poster presentation

2025

Population genetic insights of key habitat-forming coral and sponge populations in Flower Garden Banks National Marine Sanctuary

53rd Annual Benthic Ecology Meeting — Mobile, AL

Eckert RJ, Sturm AB, Pantoni GS, Herrera S, Voss JD

Oral presentation

2024

Cryptic diversity of *Stephanocoenia intersepta* corals throughout Southeast Florida and the Florida Keys

2024 AAUS Diving for Science Symposium — Fort Pierce, FL

Eckert RJ, Sturm AB, Carreiro AM, Pantoni GS, Klein AM, Voss JD

Poster presentation

2024

Cryptic diversity of *Stephanocoenia intersepta* corals throughout Southeast Florida and the Florida Keys

52nd Annual Benthic Ecology Meeting — Charleston, SC

Eckert RJ, Sturm AB, Carreiro AM, Pantoni GS, Klein AM, Voss JD

Poster presentation

2023

Genetically divergent mesophotic and shallow *Stephanocoenia intersepta* populations in the Florida Keys

51st Annual Benthic Ecology Meeting — Miami, FL

Eckert RJ, Sturm AB, Carreiro AM, Klein AM, Voss JD

Poster presentation

2023

Genetically divergent mesophotic and shallow *Stephanocoenia intersepta* populations in the Florida Keys

Mesophotic Coral Ecosystem Gordon Research Conference — Ventura, CA

Eckert RJ, Sturm AB, Carreiro AM, Klein AM, Voss JD

Poster presentation

Best overall student presentation

2023

***Stephanocoenia intersepta* corals in Florida Keys National Marine Sanctuary exhibit depth-dependent genetic structure**

Mesophotic Coral Ecosystem Gordon Research Symposium — Ventura, CA

Eckert RJ, Sturm AB, Carreiro AM, Klein AM, Voss JD

Oral presentation

2022

Genetic connectivity and algal symbiont communities of shallow and mesophotic Stephanocoenia intersepta populations in the Florida Keys

15th International Coral Reef Symposium — Bremen, Germany.

Eckert RJ, Sturm AB, Carreiro AM, Klein AM, Voss JD

Oral presentation

2022

Genetic connectivity and genetic diversity of shallow and mesophotic Stephanocoenia intersepta populations throughout Florida Keys National Marine Sanctuary.

50th Annual Benthic Ecology Meeting — Portsmouth, NH.

Eckert RJ, Sturm AB, Carreiro AM, Klein AM, Voss JD

Oral presentation

2019

Genetically distinct depth-stratified *Montastraea cavernosa* coral populations on the Belize Barrier Reef harbor relatively homogenous Symbiodiniaceae communities
48th Annual Benthic Ecology Meeting — St. John's, NL, Canada.
Eckert RJ, Reaume A, Voss JD
Oral presentation

2019

Populations of the coral species *Montastraea cavernosa* lack vertical connectivity on the Belize Barrier Reef
ASLO Aquatic Sciences Meeting — San Juan, PR.
Eckert RJ, Studivan MS, Voss JD
Poster presentation

2018

Affordable and ecologically relevant graduate research on the world's second largest barrier reef? You better Belize it!
HBOI Science Seminar — Ft. Pierce, FL.
Eckert RJ and Studivan MS
Oral presentation

2018

Connected corals: population genetics of *Montastraea cavernosa* on the Belize Barrier Reef
HBOI Ocean Discovery Center Marine Science Friday — Ft. Pierce, FL.
Eckert RJ
Oral presentation

2018

Breaking point: population genetics reveal a lack of vertical connectivity of *Montastraea cavernosa* on the Belize Barrier Reef
47th Annual Benthic Ecology Meeting — Corpus Christi, TX.
Eckert RJ, Studivan MS, Voss JD
Oral presentation

2013

Ciguatera fish poisoning in the northwestern Gulf of Mexico.
66th Gulf and Caribbean Fisheries Institute Conference — Corpus Christi, TX
Eckert RJ, Loeffler C, Robertson A, Johnston MA, Hickerson EL, Schmahl GP
Poster presentation

2009

Associational resistance and susceptibility in *Solanum carolinense*
Southeastern Ecology and Evolution Conference — Gainesville, FL.
Eckert RJ, Kim T, Underwood N
Poster presentation

FUNDING

2023

Smithsonian NMNH Research Fellowship
$16,000

2023 | 2017

FAU GPSA Travel Awards
$3,023

2021

PADI Foundation Grant
$4,377

2021 | 2019

FAU Presidential Fellowship
$10,000

2018 **FAU Thesis Award**
$500

2018 **PADI Foundation Grant**
$4,973

2009 **Bess H. Ward Honors Thesis Award**
$1,000

2009 | 2005 **Florida Bright Futures Academic Scholarship**
$14,850

2009 | 2005 **Florida State University Freshman Academic Scholarship**
$8,000

2005 **Florida State University Freshman AP Scholarship**
$500

AWARDS

2025 **Graduate Researcher of the Year**
Florida Atlantic University Northern Campus Awards

2025 **Florida Atlantic University Student Publishing Award**
Awarded for 3 publications in 2024

2021 | 2019 **Florida Atlantic University Presidential Fellow**

2020 **Florida Atlantic University Student Publishing Award**
Awarded for 3 publications in 2019

2009 | 2006 **Florida State University Dean's List**

2005 **Florida State University President's List**

CERTIFICATIONS

- **AAUS Scientific Diver**
  100 m, CCR, Trimix
- **TDI Technical SCUBA Diver**
  Advanced Nitrox, Decompression Procedures, Trimix 100 m, Closed-Circuit Rebreather, Advanced Mixed Gas CCR
- **PADI Trimix Gas Blender**
- **DAN Oxygen delivery, First Aid, and CPR for scuba diving injuries**
- **USCG Licensed Merchant Mariner**
  100 GRT
- **USFWS Motorboat Operator Certification Course**
- **Sea-Bird Electronics technician**
- **NOAA Working SCUBA Diver**

SKILLS

- **DNA and RNA extraction, PCR, qPCR, Gel electrophoresis, NGS library preparation, Agilent Bioanalyzer (DNA, RNA)**
- **Bioinformatic analyses, multivariate statistics, and data visualization**
  R statistical software, Python, Bash, PRIMER7, and CPCe

- **Coral sampling, benthic surveys, fish surveys, photography, navigation and mapping**
- **Offshore field research cruise planning, logistics, and execution**
- **Technical SCUBA diving, closed-circuit rebreather diving, operation of Nitrox compressor systems, and Trimix blending**
- **Setup, operation, and maintenance of oceanographic equipment**
  Sea-Bird Electronics, Onset HOBO, YSI
- **Adobe InDesign, Adobe Illustrator, and Adobe Photoshop**
- **Operation, navigation, and handling of small boats**
  Marlinspike, and mooring buoy construction/installation/maintenance

MEMBERSHIP AND SERVICE

Present | 2020 — **Mesophotic.org**
Content Editor

Present | 2019 — **International Coral Reef Society**

Present | 2017 — **Benthic Ecology Meeting Society**

Reviewer for:
Scientific Reports, Environmental Monitoring and Assessment, Marine Ecology Progress Series, Coral Reefs

2022 — **HBOI Graduate Student Association**
Vice President — Ft. Pierce, FL

2021, 2019 — **HBOI Graduate Student Association**
Secretary — Ft. Pierce, FL

2019 | 2018 — **Association for the Studies of Limnology and Oceanography**

REFERENCES

Dr. Joshua Voss
Ph.D. Advisor
Florida Atlantic University
jvoss2@fau.edu

Dr. Adrienne Correa
Assistant Professor
UC Berkeley
amsc@berkeley.edu

G.P. Schmahl
Superintendent
Flower Garden Banks National Marine Sanctuary
george.schmahl@noaa.gov

This curriculum vitae was made with the R package pagedown.

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